

NGUYEN HOANG THACH

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Portfolio: [hoangthach](#)

EDUCATION

Ho Chi Minh City University of Technology and Education

Bachelor of Engineering in Information Technology (GPA: 3.5/4.0)

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management, Operating Systems, Computer Networks

Ho Chi Minh City, Vietnam

September 2022 - Present

PROJECTS

Agri-Supplies E-Commerce Platform

[GitHub](#)

- Summary:** Architected and developed a robust backend system for an agricultural supply e-commerce platform, emphasizing transactional integrity and scalable business workflows.
- Architecture & Design Patterns:** Designed a transaction-safe checkout pipeline with atomic inventory allocation (FEFO/FIFO). Implemented the **State Design Pattern** to strictly govern complex order lifecycles and the **Strategy Design Pattern** to calculate dynamic product variants pricing.
- Payment & Real-time Ops:** Engineered an automated bank-transfer verification workflow via SePay integration, auto-canceling expired orders with stock rollback. Delivered instant order updates via JWT-authenticated WebSocket/STOMP channels.
- Security & RBAC:** Established a granular Role-Based Access Control (RBAC) system. Engineered an automated synchronization mechanism to dynamically scan and map API endpoint permissions into the database, streamlining authorization management.
- Procurement Module:** Built an internal supply chain module to manage suppliers, track purchase orders, and handle multi-batch inventory ingestion seamlessly.
- Tech Stack:** Java 21, Spring Boot, Spring Security, Spring Data JPA, PostgreSQL, JWT, WebSocket/STOMP.

E-Learning Platform Backend

[Live Demo](#) · [GitHub](#)

- Summary:** Architected a highly modular and scalable backend for an e-learning platform enabling course management, video processing, and secure student enrollments.
- Architecture & Patterns:** Designed a robust REST API using **NestJS** and **Prisma ORM**, strictly enforcing a Layered Architecture with the **Repository Pattern** for data abstraction. Implemented the **Factory Design Pattern** to dynamically resolve payment gateways.
- Performance & Security:** Integrated **Redis** for distributed caching to optimize frequent data retrieval. Fortified API security by implementing robust RBAC with custom Guards, JWT authentication, and **Rate Limiting** (*Throttler*) to mitigate brute-force attacks.
- Media & Real-time Ops:** Engineered a comprehensive media asset pipeline utilizing the Cloudinary API for video uploads and translation services. Built an event-driven architecture with WebSockets to deliver real-time notifications.
- Tech Stack:** NestJS, TypeScript, Prisma, PostgreSQL, Redis, WebSockets (Socket.io), JWT, Cloudinary.

BlinkyVocab - Gamified Learning Platform

[Live Demo](#) · [GitHub](#)

- Summary:** Architected a highly scalable, event-driven backend ecosystem for a gamified language learning platform, featuring interactive quizzes, achievement tracking, and community forums.
- Core Algorithm:** Engineered a personalized learning engine utilizing a hybrid algorithm combining **Spaced Repetition (SM-2)** and **Forgetting Curve** models to dynamically calculate and optimize vocabulary review schedules.
- Architecture & Performance:** Designed a modular REST API with **NestJS** and **Prisma ORM**, strictly adhering to the **Repository Pattern**. Integrated **Redis** for robust distributed caching and rate-limiting to ensure high performance.
- Real-time & AI Integration:** Built a centralized, event-driven **WebSocket** architecture (*RealtimeDispatcher*) to seamlessly broadcast instant notifications and live community interactions. Integrated external AI services for automated vocabulary context generation.
- Security Governance:** Fortified system security with a multi-layered authentication strategy combining custom Guards, JWT, Google OAuth2 workflows, and granular Role-Based Access Control (RBAC).
- Tech Stack:** NestJS, TypeScript, Prisma, PostgreSQL, Redis, WebSockets, JWT, OAuth2, AI APIs.

SKILLS

Programming Languages:	Java, SQL, JavaScript/TypeScript, Python
Backend Frameworks:	Spring Boot, Node.js (Express.js, NestJS)
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Tools & Cloud Services:	Git, Docker, REST APIs, WebSocket/STOMP, JWT, Cloudinary
Languages:	English (TOEIC 600)

ACTIVITIES & AWARDS

Winner, 'Best Idea Award' - HCMUTE Hackathon 2025

Spring 2025

- Awarded for developing 'TâmTrò', a full-stack web application designed to provide 24/7 AI-powered mental health support for users.
- Engineered the backend RESTful API using Node.js and Express.js to manage user authentication, real-time chat sessions, and mental health assessment data.
- Implemented the core emotion analysis feature, processing chat history to generate and visualize a user's emotional trends over time.